ADA — Palm Oil Retention Architecture

Program Version 5.4 (Prototype A) for the Nebico Melting and Retention System

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Operation, Architecture, Implemented Changes and Resolved Defects of the
Adaptive Demand Allocation Ladder Program on the Coolmay L10S

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This document describes the architecture of Program 5.4, the first revision of the ADA ladder program in which the complete chain operates on the machine: temperature and level sensing, thermal demand registration, three-stage electric heating, solar integration, and multi-station oil dispensing over Modbus. It records how the architecture differs from the 3.x series, the design decisions behind the simplification to roughly 1,800 steps, and the defects corrected in this revision.

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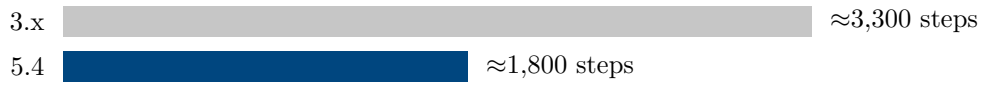
1 Summary

The Nebico system stores palm oil, keeps it warm enough to dispense, and delivers it to three stations, all from one ladder program on a Coolmay L10S. Version 5.4 (Prototype A) is the first revision in which the whole chain works on the machine.

≈1,800	27 kW	4 + 3	6 + 4
program steps	heating, 3 stages	tanks + dispense stations	RTDs + level sensors

Version 5.4 at a glance

- Sensing, heating, solar and Modbus oil dispensing working together
- Demands registered and acted on as they arise, with no priority scoring
- Prototype A chosen for Modbus polling and carried forward



What the system can do

- Stores palm oil in a primary tank and three secondary tanks
- Keeps the oil liquid and dispensable
- Heats water automatically, with heating safety
- Keeps the primary tank filled with the AODD pump
- Uses solar heat to save electricity
- Dispenses oil at three stations

Basic operation at the front panel

- Three rows: emergency and mode (row 1), equipment (row 2) and the three heaters (row 3)
- A lit indicator shows that the function is active
- Oil: turn the station selector on, then use the input switch; turning the selector off stops the oil line in an emergency
- Full details are in Section 3

Concerning things

- Switching the system off quickly can cause water hammer
- The heater contactor may chatter at a particular water level
- Overfilling of the heating tank is still being monitored
- Measured oil dispensing is experimental, and flow may change with pressure at distant stations
- Control-panel wiring and connectors are untidy
- Details are in Section 5

2 What the System Achieves

1. **Storage of palm oil in primary and secondary tanks.** The primary tank (T5, the 5 kL holding and replenishment tank) and the secondary tanks (TA, TB and TC, 10 kL each) hold the oil under level monitoring, with level, percentage and mass calculated for every tank.

2. **Oil kept liquid and dispensable.** Submerged hydronic coils keep the palm oil above its melting range. Each tank registers a heat need when its temperature falls below its threshold, and only the branch gates of the tanks that need heat are opened. Oil can then be dispensed to three stations through the gear pump.

3. **Automatic water heating with heating safety.** The heater target is chosen automatically from the primary tank level, and the three 9 kW heater stages are staged by the gap between target and actual temperature, with 10 s holds against flicker. The heaters are cut off on low water, on E-stop, and outside auto mode unless switched manually. A blended-temperature check limits T5 to 65 °C (to be lowered to 55 °C).

4. **Continuous filling of the primary tank.** The AODD pump starts when the primary tank falls below 40 % and stops at 60 % (currently limited while tank calibration is completed, with a target of 90 %), so the primary tank is kept supplied without operator action.

5. **Solar integration to save electricity.** The solar cycle runs when the tank water is colder than the collector loop, so that solar heat displaces electric heating. In this revision the solar cycle is held closed for overfill monitoring (Section 5).

The limitations that go with these capabilities are listed in Section 5.

3 Operation Manual

3.1 Front Panel

The front panel has three rows of paired indicators and switches: two positions in the first row, five in the second and three in the third. An indicator glows when its function is active.

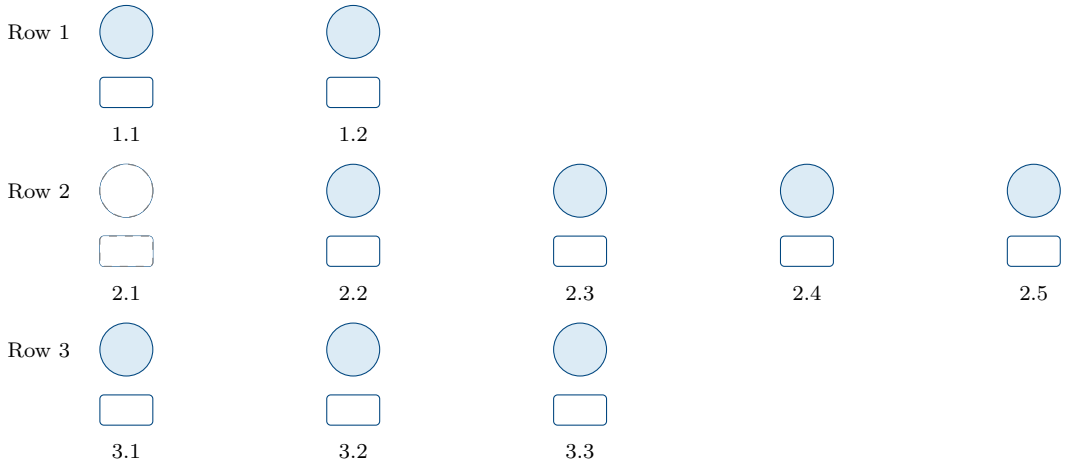


Figure 1: Front panel layout (indicator above, switch below; dashed = unused).

Table 1: Front panel indicators and switches.

Pos.	Row	Indicator (if glowing)	Switch
1.1	R1	Emergency tripped	E-Switch
1.2	R1	Automation mode	Auto/Manual mode
2.1	R2	none	none
2.2	R2	AODD pump on	AODD switch
2.3	R2	Solar cycle on	Solar switch
2.4	R2	Oil pump on	Oil switch
2.5	R2	Circulation pump on lowest setting	Circulation switch
3.1	R3	Heater 1 on	Heater 1 switch
3.2	R3	Heater 2 on	Heater 2 switch
3.3	R3	Heater 3 on	Heater 3 switch

In auto mode the program decides which equipment runs. In manual mode the row 2 and row 3 switches operate the corresponding equipment directly.

3.2 Oil Panel

Each station has an oil panel with two switches. Their switch positions are undetermined and are to be corrected by the field engineer.

Selector switch.
It must be turned on before any processing, and it is also the emergency switch in case of a mishap: simply turn the selector off.

Input switch.
It starts the pumping, as described below.

Once the selector is on, there are two possibilities.

Manual mode.
The value at the HMI is not read, regardless of what it is. Turning the input switch on starts pumping the liquid until the switch is turned off again.

Auto mode.
If the HMI input is kept at 0, the panel behaves as in manual mode, which is kept for priming of the pipe. Entering a value and turning the input switch on dispenses the set amount of liquid. The physical condition of the input switch does not matter once it is on: dispensing continues until the amount has been dispensed or the selector is turned off. To dispense again, turn the input switch off and on.

3.3 Suggested Upgrade

Suggestion from the author. The input switch should be a momentary switch. Priming then requires a continuous press, and a measured dispense can be repeated with a simple click each time. The three-way switch should be replaced with a two-way switch. A minor, non-essential improvement is to arrange the switches in the same way at every station so that all stations look identical to one another.

4 Architecture

This section describes the program function by function, in the order in which the logic works through the machine. The register-level detail of each function (tables and equations) follows its description.

Disclaimer on program steps. The step numbers quoted in this document may not be exact. The function described will be similar to, or located around, the given point in the ladder, so the numbers should be used to find a block and not as precise addresses. The source is the ladder export of project `MAIN` dated 29 September 2026 together with the design decisions confirmed by the author.

4.1 System Overview

The machine keeps palm oil above its melting range in storage and dispenses it to filling stations on request.

- Three 10 kL retention tanks (TA, TB, TC) and one 5 kL holding and replenishment tank (T5) keep the oil warm through submerged hydronic coils.
- Heat is delivered by a water loop. A three-stage electric heater bank of 9 kW per stage supplies the loop, and a solar loop is integrated alongside it.
- Four branch gates route hot water to whichever tanks need heat. The circulation pump speed follows the number of open gates.
- Water make-up is automatic, and the heaters are cut off if the water level falls below the safety level.
- Oil is dispensed to three stations through a gear pump. Each station enters a requested mass on its own HMI, and the PLC polls that value over Modbus.

4.1.1 Controller and instrumentation

The program runs on a Coolmay L10S and is roughly 1,800 steps long (the `END` instruction is at step 1796). Six RTD temperature sensors and four level sensors feed the analog inputs. The RTD assignment is fixed as follows.

Table 2: RTD assignment.

Register	Location	Remarks
D34	T5 holding tank	Also used in the T5 need rung and the blend check
D36	TA	
D38	TB	
D40	TC	
D122	Solar loop	Collector side of the solar comparison
D42	Heater / water	Heater feedback, solar comparison, blend check and dispense correction

Register D10 (source D8231) is read but not used elsewhere. It is reserved for a flow meter to be added later.

4.1.2 Architecture at a glance

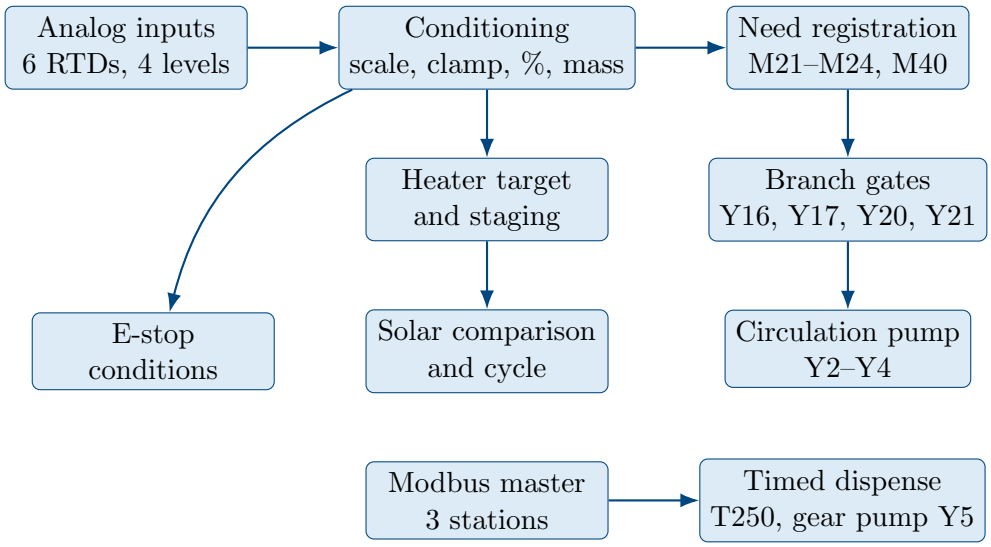


Figure 2: Functional structure of Program 5.4.

4.1.3 Program block map

Table 3 lists the blocks in the order in which they appear in the ladder. Where the export label did not match the function of the block, the function is stated in the last column.

Table 3: Program blocks of the 5.4 ladder.

Step	Export label	Function
0	Timers	Power-on delays T0–T3 (1, 2, 3, 5 s) releasing later blocks in stages; T50 is the 5 s sampling pulse
16	Statement	Start-up pulse (M8002): serial port and Modbus set-up writes
44	Analog Signals	Reads all analog channels and converts to float
133	Sensor Logging/Calibration	One linear rung per sensor: temperatures ×0.1, levels gain and offset
370	Level Correction	Negative level replaced by 0
415	Corrected Level	Level as percentage of tank height
520	Level to Mass	Level to kg, with the same zero clamp
626	Heater Latch	Selects the heater target temperature, banded by T5 level
722	Heater Gate	Target-minus-current gap split into four bands M15–M18
815	Heater Latch	10 s hold timers on each band (anti-flicker)
862	T5 Max Temperature	Blended temperature check against 65 °C
947	E-Stop	E-stop trip conditions
1015	Automation	Auto mode latch M20
1020	Need Scheduler	Integer conversions and per-tank need flags
1142	Solenoid Gating	Counts how many gates are demanded (M25–M27)
1217	Heater Gate (Outputs)	Drives heaters Y10–Y12 from the bands
1244	Solenoid Gates	Drives branch gates Y16, Y17, Y20, Y21
1272	Solar Gating	Solar cycle decision and outputs
1297	Manual Solenoid Gate	Manual gate count M60–M62
1372	Circulation Pump Gates	Speed steps Y2–Y4
1403	AODD Pump Gates	AODD pump Y7
1418	HMI Chart	Copies inputs X0–X37 to M115–M146
1482	Auto-Fill	Water make-up valve Y14
1486	Oil Line Auto Mechanics	Starts the Modbus polling cycle
1506	MODBUS Lines	Round-robin poll of three stations
1582	MODBUS Calculation	Selected station value to dispense time
1707	Oil Gate Schedule	Station selection, first come first served
1719	Oil Gate	Oil-line gates Y22–Y24
1728	Oil Auto-Dispense (Timer)	Runs and ends the timed dispense
1762	Gear Pump	Gear pump Y5
1778	MODBUS Debugger	Copies the comm error code to D700
1784	Oil Line disconnect	Clears “done” and “running” when the selector is released

4.2 Sensor Acquisition, Calibration and Derived Math

Sensor acquisition.
A 5 s timer triggers the readings, and a 5 s gap is kept between recordings to smooth the curve.

Sensor calibration.
The sensors are lightly calibrated. Temperature is divided by 10, whereas the level is limited between the empty tank and the full tank. A tank registering below 0 is automatically adjusted to 0.

Derived math.
Level and mass are calculated from the conditioned readings.

4.2.1 Sampling

The timer T50 (K50 on a 100 ms base, that is 5 s) fires the read block and is reset at step 316. Values are not averaged, so they follow the sensor. The sampling interval of 5 s is intended.

Table 4: Analog channels and scaled results.

Raw	Source	Scaled to
D0, D1	Onboard analog (D8030, D8031)	D34, D36: temperature, ×0.1
D2–D5	Special block 2, four channels	D38, D40, D122, D42: temperature, ×0.1
D6–D9	Special block 1, four channels	Level (mm) after gain and offset
D10	D8231	D32: read only, reserved for a flow meter

4.2.2 Level calibration

Each level sensor has a single linear rung. The tank height sets the percentage and the density constant sets the mass.

Table 5: Level, percentage and mass conversion per tank.

Tank	Level (mm)	Height for %	Mass per mm
T5	raw × 0.154 − 2800	2000 mm	1.95 kg/mm
TA, TB, TC	raw × 0.2246 − 3623	2760 mm	3.3 kg/mm

Results are held in D52–D58 (level), D76–D82 (percent) and D84–D90 (mass).

4.2.3 Zero clamp

A clamp is applied at each stage. If the level is not positive it becomes zero, and so do the percentage and the mass derived from it. The clamp rung uses the internal relay M0, which is lit when the level is positive. The contact used in the clamp rung is normally closed, so a non-positive level falls through to the zero replacement.

4.3 Water Temperature Target and Heater

Water temperature target.
The target is 40, 50 or 60 °C, depending on the level of the holding tank.

Heater.
The number of heaters switched on depends on the gap between the target and the current temperature: a larger gap opens more heaters, and a smaller gap fewer. Each state latches automatically with a minimum timer of 10 s, which avoids rapid heater on/off switching during operation.

Maximum T5 temperature.
This keeps the tank temperature from ever passing the set limit (currently 65 °C, to be adjusted to 55 °C).

4.3.1 Heater target (step 626)

The target D200 is chosen automatically from the level of the holding tank (T5) and takes one of three values: 40, 50 or 60 °C. When M29 is set, the HMI value D202 overrides the banded target. M29 is the manual target-temperature setting.

4.3.2 Heater staging (steps 722, 815, 1217)

The gap is the target minus the current temperature (integer, in °C):

$$D96 = D200 - D94. \tag{1}$$

It is split into four bands, and each band selects the number of heaters that run. Y10, Y11 and Y12 are water heaters of 9 kW each, so the stages correspond to 9, 18 and 27 kW.

Table 6: Heater bands and stages.

Band	Gap (°C)	Heaters on	Power
M15	0 to 3	Y12	9 kW
M16	3 to 6	Y12, Y11	18 kW
M17	6 or more	Y12, Y11, Y10	27 kW
M18	0 or below	none	0

Each band is latched and held for at least 10 s (T30–T33, K100) before it can change, so that the heaters do not flicker. Heater outputs also require auto mode, no E-stop, and the water-level interlock on X11. X11 is the input that indicates the water level is too low, and it cuts the heaters off. In manual mode X6, X7 and X10 switch Y10, Y11 and Y12 directly.

4.3.3 T5 maximum temperature (step 862)

A blended temperature is computed as

$$T_f = \frac{m_{T5}\,T_{T5} + 239.53\,T_{D42}}{239.53 + m_{T5}}, \tag{2}$$

where m_{T5} is D84 (kg), T_{T5} is D34 and T_{D42} is D42, the heater/water RTD. If the integer of T_f exceeds 65 °C, the flag M19 is set. M19 acts on the T5 need rung (M21): it stops the scheduling so that circulation to T5 is cut off while the limit is exceeded.

The limit is currently 65 °C and needs to be adjusted to 55 °C.

4.4 Core Schedule, Solenoid Gates and Pump Speed

Core schedule.
It schedules the heat supply as needed, and at the same time monitors the T5 level and supply.

Solenoid gates.
All gates are normally closed. In manual mode they are open until closed from the HMI, and in auto mode they are decided by the core schedule.

Pump speed.
The pump speed adjusts itself to the number of gates opened, in all modes.

4.4.1 Need registration and gates (steps 1020–1265)

Each tank registers a need when its temperature is below a threshold.

Table 7: Need flags, thresholds and gate outputs.

Need	Tank	Sensor	Threshold	Gate
M21	T5	D34	below 35 °C	Y21
M22	TA	D36	below 30 °C	Y20
M23	TB	D38	below 30 °C	Y17
M24	TC	D40	below 30 °C	Y16

Gates open as needed in any combination, with no priority. Manual gates M500–M503 from the HMI drive the same outputs. When the physical E-stop X0 is active, all four gates are forced open.

The gate-count flags select the circulation pump speed.

Table 8: Gate count and pump speed.

Flag	Condition	Pump output
M25	exactly one need	Y2
M26	exactly two needs	Y3
M27	three or four needs	Y4

In manual mode M60–M62 do the same for M500–M503, enabled by X2.

4.5 Solar Gate

The solar and heater gates are opposite to each other, and both cannot be closed at the same time. In auto mode, if the solar temperature is higher than the current temperature, the system gives priority to transferring water between the solar loop and the heater so

that heat is exchanged. When the pump is off, the valves close automatically to prevent water from trickling back through the pipes, which was observed on site and caused the overfill.

4.5.1 Solar loop (step 1272)

The solar cycle runs when the tank water is colder than the collector loop (comparison of D120 with D124, flag M28). Two valves take part: opening Y6 starts the solar cycle, and opening Y15 puts the loop back on itself. Y6 and Y15 follow M28 in auto mode and X4 in manual mode. When pump is turned off, both valves are closed to prevent the water from the loop to trickle back into the system and potentially overfill it.

4.6 AODD Pump and Auto-Fill

AODD pump.
It opens only if the tank is below 40 % and stops once the tank reaches 60 %. The stop level should be 90 % but is kept lower for safety, to avoid overflow. Once the tank is above 60 % the pump cannot be turned on, even manually.

Auto-fill.
By design, water flushes back from the solar loop to the heating tank only when the solar loop is full. Monitoring the level in the heating tank and gating the refill tank therefore keeps both tanks in check. Auto-fill works in tandem with the level sensor inside the heating tank, which prevents the heater from turning on once the level is below the threshold.

4.6.1 Water make-up and AODD pump

- **Auto-fill.** X11 (water level too low) sets the make-up valve Y14 and X12 (water level stable high) resets it. The heaters are cut off if water falls below the safety level.
- **AODD pump Y7.** In auto mode it starts when T5 is below 40 % (M40) and stops at 60 % (D128 $\geq$ 60). The cap is 60 % rather than 90 % while the tank is still being calibrated. In manual mode X5 runs it directly, but it cannot be started once the tank is above 60 %.

4.7 E-Stop and HMI Chart

4.7.1 E-stop conditions (step 947)

The E-stop output Y0 is the error indicator. It asserts on either of the following:

- any oil tank TA, TB or TC above 90 %; or
- the T5 holding tank below 20 % (empty).

The E-stop lamp lights and the HMI beeps. The physical E-stop X0 also forces all gates open and drops the heaters and manual pumps. The flag M400 overwrites Y0 so that the emergency indication can be silenced when necessary.

4.7.2 Loss of power

On loss of power everything powers down. The gates close and the instruments sleep. This is the reason for choosing normally closed gate valves for the branch circuits.

4.7.3 Removed emergency monitoring

Emergency monitoring of falling or rising sensor readings is not present in 5.4. It was dropped completely because it was judged unnecessary in the field. The only comm-failure handling visible in the ladder is the 0.5 s timeout advance in the Modbus poll (Section 4.8).

4.7.4 HMI chart

The input and output ports (X and Y) are displayed on the HMI so that faults can be monitored when the panel is closed and observing the lamps is not practical. The inputs X0–X37 are mirrored to M115–M146 (around step 1418).

4.8 Modbus Communication, Oil Gate and Oil Dispense

Modbus communication.
Modbus is polled only in auto mode. The loop continuously scans the three HMIs for a numerical input and attaches each value to its designated data register.

Oil gate.
Oil gating is first come, first served: when one gate is open, the others cannot be open. The gear pump of the oil line is tied to the oil gate, so if no oil gate is selected the oil pump stops, regardless of the operation mode.

Oil dispense.
The dispensed amount is calculated from a calibrated flow rate (to be replaced by a flow meter) at a given temperature. A temperature adjustment, calculated manually first, then corrects for the different viscosity of the oil at other temperatures. Finally the result is calibrated so that the dispensed amount better matches the mass entered.

4.8.1 Polling (steps 1486–1582)

Polling runs only in auto mode (M20). The PLC is master and reads one holding register from each of three stations in round robin. Register H1 at each station holds the required mass entered at that station.

Table 9: Modbus round-robin poll.

Order	Slave	Function	Address	Stored in	Selector
1	2	H03	H1	D400	X22
2	1	H03	H1	D402	X24
3	3	H03	H1	D404	X26

Each step waits 0.5 s (T10, T12, T14) before the read. It advances when the instruction completes (M8029) or after a 0.5 s timeout (T11, T13, T15), so an unresponsive station cannot block the others. The comm error code is copied to D700 by the debugger block.

4.8.2 Oil-line gates

The oil-line valves Y22–Y24 are normally open valves, so an energised output means the line is closed. Each valve is driven by the selectors of the *other* two stations. The operating procedure for the station is given in Section 3.2.

4.8.3 Dispense timing (steps 1582–1750)

The polled value of the selected station is converted to float (D180) and turned into a run time:

$$t = \frac{M_{\text{req}}\, D360}{D356\, \exp\!\big(1.0\,(30 - T_{D42})\big)} \quad [\text{s}], \tag{3}$$

$$D196 = \text{INT}(10\,t) \quad [100\,\text{ms ticks}], \tag{4}$$

where $M_{\text{req}} = D180$. The retentive timer T250 (100 ms) runs for D196 ticks with the gear pump on. At timeout it clears D194 and D196, resets T250 and sets the “done” flag M451.

Calibration completed for intial use. The constants for flowrate (0.4) and temperature coefficient (0.0084) has been calibrated crudely. Regardless the dispensed value is closing in at Error rate of 0.1% for 28kg of dispensed palm oil.

5 Concerns

Prototype status. The system is a prototype and is open to upgrades. The points below describe present limitations and what a technician should keep in mind. They are not defects in the program logic.

Water hammer on a quick shutdown.
The VFDs turn off and on slowly. When the system is turned off in a short manner, the chance of water hammer on the normally closed solenoid valves increases. The water pressure from the VFD is limited even in its high-pressure mode, but this remains a point of concern, to be kept in mind by the technician if the system later requires maintenance. It is a drawback of the arrangement; normally closed valves nevertheless prevent water from flushing backwards.

Overfilling of the heating tank.
Overfilling is still open, despite our engineering conclusion that the overfill was due to the return of the water from the circulation pipe network and implementing change in the software, it must be monitored closely.

Heater chatter at a particular water level.
The heater timeout and sequence were not planned for the water level of the tank rising and falling. At a particular level the contactor may switch on and off rapidly. This follows from the non-latching design, which was removed to keep the program light. If problems are seen downstream, the PLC technician or engineer may gate the heater, with an advised 10 s.

Oil flow at distant nodes.
Oil distributed to several nodes at different distances may make the gear pump flow rate change as the pressure increases (and possibly decreases). The operational program does not account for this, so changes would have to be made in the oil calculation rungs. This needs calibration and PLC programming at a higher skill level, and contacting the author is advised.

Wiring and connectors.
The wires are routed loosely, and the connectors in the control panel are still messy. This should be addressed before handover.

6 Architectural Changes from the 3.x Series

The 5.4 program is a deliberate simplification. The earlier 3.x series used priority and scoring-based schedulers and averaged sensor values. In practice these cost program size and general registers without a matching benefit on the machine. Version 5.4 registers a demand when it arises and acts on it, with no ranking between demands.

Table 10: Comparison of the earlier 3.x revisions with 5.4.

Area	Earlier (3.x)	5.4
Program size	about 3,300 steps	about 1,800 steps
Scheduling	Priority and scoring-based schedulers	No priorities; a demand is registered and fired as it arises
Analog signals	Averaged	Instantaneous values, sampled every 5 s; saves general registers
Calibration	Several stages	One rung per sensor; negatives clamped to 0 (level, %, mass)
Heater target	Math-based	Three fixed bands by T5 level, no math
Heater staging	Not banded	Gap-based stages with 10 s holds
T5 maximum	No dedicated limit	65 °C, the highest expected in the field; to be adjusted to 55 °C
Branch gates	Normally open assumed	Normally closed, open on demand; all open in emergency
Jacket pump	Separate	Merged into the circulation pump
Oil line	Scheduler	Selector-based, first come first served, with Modbus polling
Emergency monitoring	Included rising and falling sensor-reading checks	Dropped as a whole; deemed unnecessary in the field

6.1 Rationale

Demand registration in place of scheduling.
A tank whose temperature is below its threshold simply registers a need, and its gate opens. Gates may open in any combination. Removing the score and priority arithmetic removed most of the register pressure of the 3.x series.

Instantaneous analog values.
The values are sampled once per 5 s pulse and are not averaged. This is intended: the process is slow relative to the sampling interval, and the saving in general registers was worth more than the smoothing.

Fixed heater bands.
The heater target no longer follows a formula. It takes one of three fixed values chosen by T5 level.

A dedicated T5 limit.
65 °C sounds a bit high and is considerably high but as of the inital monitoring by the author, the program over-estimates the heating power of the water coil. The increase is in rationale to the findings from the site and delierate by the author. The real expected maximum temperature is the highest at 55 °C

Normally closed gates.
The branch gates are supplied normally closed and open on demand. On loss of power they close, so a power failure never leaves water circulating unattended.

Merged pump.
The jacket pump is folded into the circulation pump, whose speed follows the number of open gates.

Emergency monitoring reduced.
Rising and falling sensor-reading checks were dropped completely, because they were judged unnecessary at the field. The remaining emergency conditions are the level-based trips in Section 4.7.1.

7 Defects Corrected in Version 5.4

Table 11: Defects and clean-ups in this revision.

Item	Defect	Correction
Station 3 auto-dispense (step ≈1735)	The rung referenced M303, which is not defined anywhere. As read, station 3 auto-dispense would never start.	Corrected to M302.
Dispense-running flag M450	The point at which M450 (dispense running) was reset could not be found in the ladder, so re-arming after "done" was unclear.	M450 is now reset in the oil disconnect line, in addition to M451 at step 1761.
M200 and M202 (Oil Line Auto Mechanics)	Redundant relays with no function.	Removed.
Block labels	Blocks at steps 947, 1020 and 1217 carried misleading labels.	Correct functions recorded in Table 3; relabelling suggested.

Effect of the M302 correction. Y22–Y24 are driven by the complementary selectors, so the undefined M303 meant that the line-blocking and start conditions for station 3 never became true. With M302 in place the station 3 chain follows the same pattern as stations 1 and 2.

8 Field Status and Limits

- Modbus polling works on the machine, and Prototype A is complete.
- Oil-line dispensing performs crudely and is operational but not in its final form.
- The AODD pump is limited to 60 % until tank calibration is finished (target 90 %).

- T5 maximum is 55 °C, the highest temperature expected in the field; it is to be adjusted to 55 °C.
- Register D10 is reserved for a flow meter to be added later.
- The start-up serial and Modbus set-up writes (D8400, D8401, D8114, M8192, D8397 and D204) are required for the controller to operate and are retained unchanged, although their individual purposes have not been established.

9 I/O Reference

Table 12: Inputs.

Input	Function
X0	E-stop
X1	Auto mode
X2	Manual circulation pump
X4	Manual solar cycle
X5	Manual AODD pump
X6, X7, X10	Manual heater stages (Y10, Y11, Y12)
X11	Water level too low (heater interlock, auto-fill set)
X12	Water level stable high (auto-fill reset)
X21, X23, X25	Station dispense start
X22, X24, X26	Station select
X0–X37	Mirrored to M115–M146 for the HMI

Table 13: Outputs.

Output	Function
Y0	E-stop / error indicator (overridable by M400)
Y1	Auto lamp
Y2, Y3, Y4	Circulation pump speed 1, 2, 3
Y5	Gear pump
Y6	Solar cycle valve (open = solar cycle); forced closed in 5.4
Y15	Loop-back valve (open = loop back)
Y7	AODD pump
Y10, Y11, Y12	Heater stages, 9 kW each
Y14	Auto-fill valve
Y16, Y17, Y20, Y21	Branch gates (M24, M23, M22, M21)
Y22, Y23, Y24	Oil-line gates (normally open, energised = closed)